(B) The coulomb force between two point charges is 1.5×10^{-4} N. How will this force be affected in the following situations :

(i) the distance between the charges is doubled.

(ii) the magnitude of each charge is doubled but the distance between the charges remains the same.

(B) The coulomb force between two point charges is 1.5×10^{-4} N. How will this force be affected in the following situations :

(i) the distance between the charges is doubled.

(ii) the magnitude of each charge is doubled but the distance between the charges remains the same.

(b)	(i) Force becomes one forth	1
	(ii) Force becomes four times	1

Two identical conducting balls A and B have charges –Q and +3Q respectively. They are brought in contact with each other and then separated by a distance d apart. Find the nature of the Coulomb force between them.

Two identical conducting balls A and B have charges –Q and +3Q respectively. They are brought in contact with each other and then separated by a distance d apart. Find the nature of the Coulomb force between them.

Repulsive | 1

A small sphere S_1 with charge $-8q$ is 1·6 m away from another identical sphere S_2 with charge $+2q$. The two spheres are brought in contact with each other and then separated by a distance 1·6 m. Initially the force between the two spheres was $8 \cdot 1 \times 10^{-4}$ N.

Based on the above facts, answer the following questions :

(i) Which sphere will transfer the electrons to the other sphere after they were brought in contact ? *1*

(ii) How does the net electric field at the midpoint on the line joining the two spheres change after contact ? *1*

(iii) What was initial charge on spheres S_1 and S_2 ? *2*

OR

(iii) What is the charge on spheres S_1 and S_2 after contact ? *2*

(i) Identification of sphere	1
(ii) Effect on net electric field after contact	1
(iii) Initial charge on S_1 and S_2	1+1
OR	
Charges on S_1 and S_2 after contact	1+1

(i) Electrons will be transferred from sphere S_1 to sphere S_2 when brought in contact. 1

(ii) net electric field changes from non zero to zero at the midpoint of line joining the centres of the two spheres. 1

(iii) $$\frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2} = F$$ ½

$$9\times10^9 \times \frac{(8q)(2q)}{1.6\times1.6} = 8.1\times10^{-4}\,N$$

$$16q^2 = \frac{8.1\times1.6\times1.6\times10^{-4}}{9\times10^9}$$ ½

$$q^2 = \frac{8.1\times1.6\times1.6\times10^{-4}}{9\times10^9\times16}$$

$$q^2 = 9\times16\times10^{-16}$$

$$q = 12\times10^{-8}\,C$$ ½

Charge on S_1 = $-8q = -96\times10^{-8}$ C

Charge on S_2 = $2q = 24\times10^{-8}$ C ½

OR

(iii) Charges on S_1 and S_2 after were brought in contact

$-3q$ and $-3q$ 2

4. A particle of mass m and charge –q is moving with a uniform speed v in a circle of radius r, with another charge q at the centre of the circle. The value of r is :

(a) $\frac{1}{4\pi\varepsilon_0 m}\left(\frac{q}{v}\right)$

(b) $\frac{1}{4\pi\varepsilon_0 m}\left(\frac{q}{v}\right)^2$

(c) $\frac{m}{4\pi\varepsilon_0}\left(\frac{q}{v}\right)$

(d) $\frac{m}{4\pi\varepsilon_0}\left(\frac{q}{v}\right)^2$

4. A particle of mass m and charge –q is moving with a uniform speed v in a circle of radius r, with another charge q at the centre of the circle. The value of r is :

(a) $\dfrac{1}{4\pi\varepsilon_0 m}\left(\dfrac{q}{v}\right)$

(b) $\dfrac{1}{4\pi\varepsilon_0 m}\left(\dfrac{q}{v}\right)^2$

(c) $\dfrac{m}{4\pi\varepsilon_0}\left(\dfrac{q}{v}\right)$

(d) $\dfrac{m}{4\pi\varepsilon_0}\left(\dfrac{q}{v}\right)^2$

(b) $\dfrac{1}{4\pi\varepsilon_0 m}\cdot\left(\dfrac{q}{v}\right)^2$

32. (a) (i) State Coulomb's law in electrostatics and write it in vector form, for two charges.

(iii) Two charges A (charge q) and B (charge 2q) are located at points (0, 0) and (a, a) respectively. Let $\hat{i}$ and $\hat{j}$ be the unit vectors along x-axis and y-axis respectively. Find the force exerted by A on B, in terms of $\hat{i}$ and $\hat{j}$. *5*

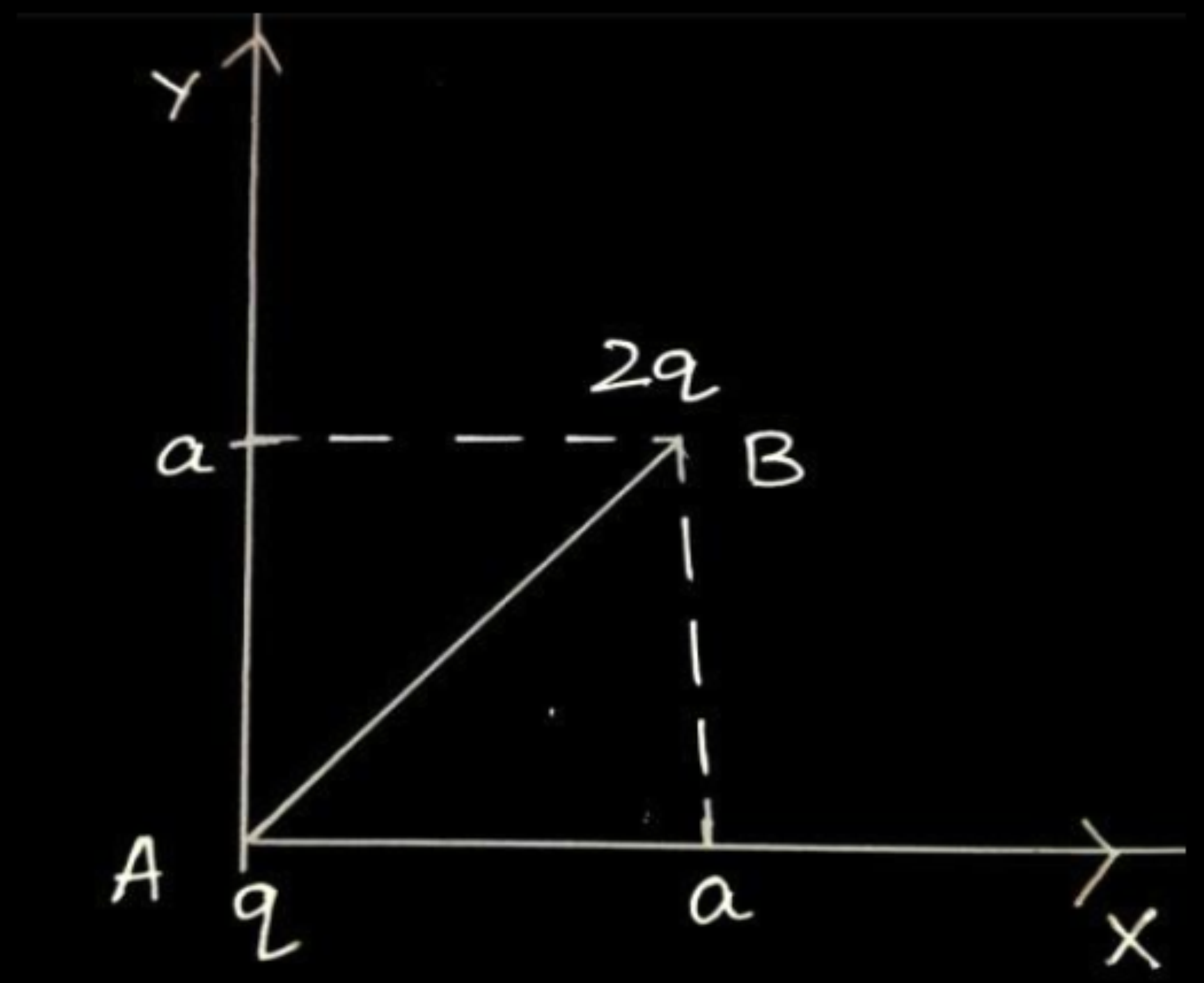
i) Force between two point charges varies inversely with the square of distance between the charges and is directly proportional to the product of magnitude of the two charges and acts along the line joining the two charges.

$$\overrightarrow{F_{12}} = \frac{1}{4\pi \in_o} \frac{q_1 q_2}{r_{12}^{\ 2}} \widehat{r_{12}}$$

1

1

iii)



$$\vec{r} = \overrightarrow{AB} = a\hat{i} + a\,j$$

r = $\left|\overrightarrow{AB}\right| = \sqrt{a^2 + a^2} = \sqrt{2}a$ 1

$$\vec{F} = \frac{1}{4\pi \in_o} \frac{q_1 q_2}{r^2} \hat{r}$$ 1

1 $$\vec{F} = \frac{1}{4\pi \in_o} \times \frac{q \times 2q}{(\sqrt{2}a)^2} \times \frac{(a\hat{i} + a\,j)}{\sqrt{2}a}$$

$$\vec{F} = \frac{1}{4\pi \in_o} \times \frac{2q^2}{2a^2} \times \frac{(\hat{i} + j)}{\sqrt{2}}$$

$$\vec{F} = \frac{1}{4\pi \in_o} \times \frac{q^2}{\sqrt{2}a^2} \times (\hat{i} + j)$$

½ $$\vec{F} = \frac{q^2}{4\sqrt{2}\pi \in_o a^2} (\hat{i} + j)$$

Note: Award 1 mark if a student calculates the magnitude of force only.

$$\left|\vec{F}\right| = \frac{1}{4\pi \in_o} \frac{q^2}{a^2}$$

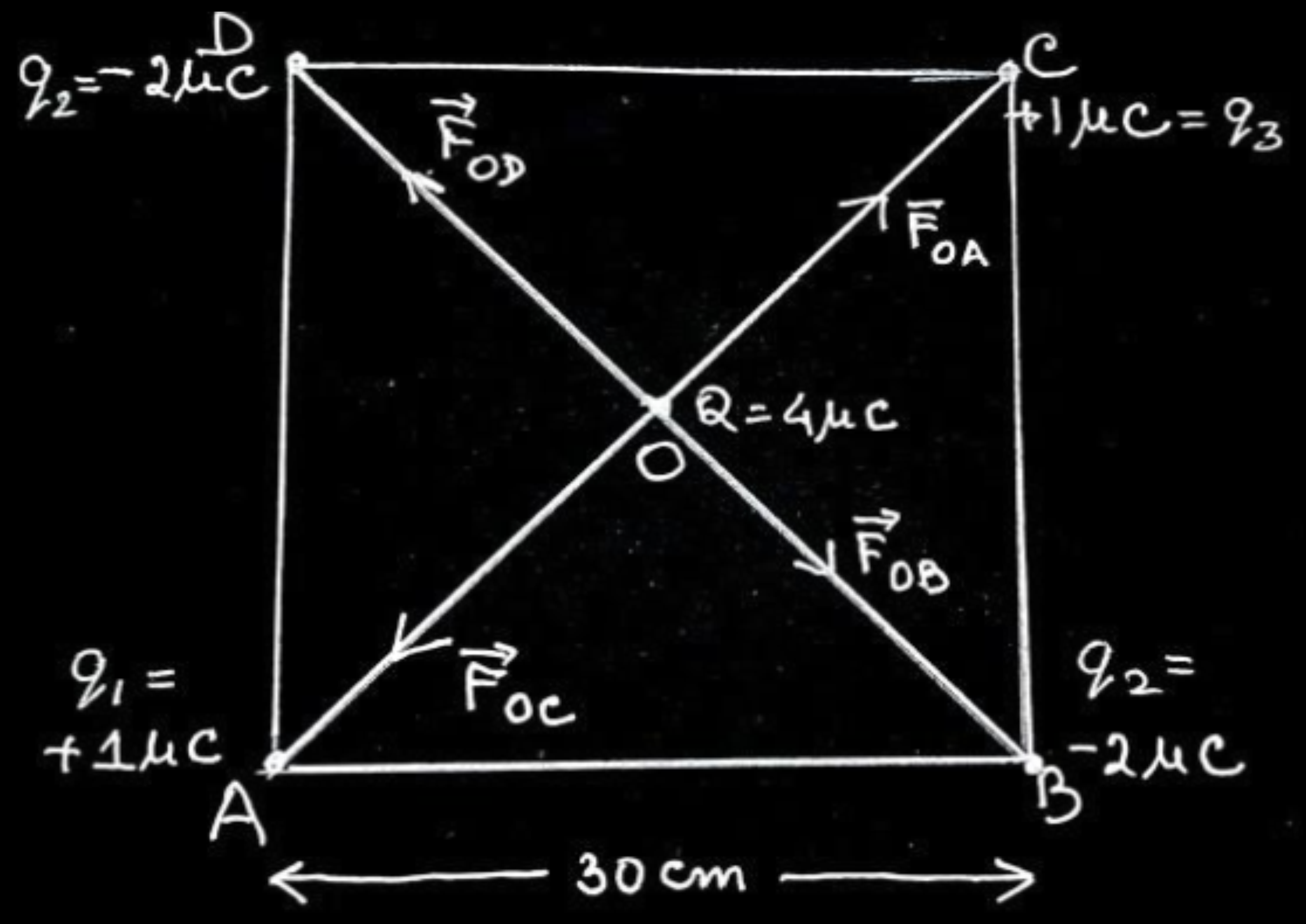
Alternatively

Give full credit if a student uses component method to solve the question.

(iii) Two point charges Q_1 (40 μC) and Q_2 (-16 μC) are placed along x-axis at 0 cm and 24 cm from the origin respectively. Calculate the net force on a third charge Q_3 ($-$ 2·5 μC) placed at x = 36 cm from the origin.

(iii) $F = F_2 - F_1$	
$= \frac{kQ_2Q_3}{{r_{23}}^2} - \frac{kQ_1Q_3}{{r_{13}}^2}$	1
$F = \frac{9\times10^9 \times 16\times 2.5\times10^{-12}}{(12\times10^{-2})^2} - \frac{9\times10^9 \times 40\times 2.5\times10^{-12}}{(36\times10^{-2})^2}$	
$F = 25 - 6.94$	
$F = 18.06\,N$	1

18. (a) Four point charges of 1 μC, –2 μC, 1 μC and –2 μC are placed at the corners A, B, C and D respectively, of a square of side 30 cm. Find the net force acting on a charge of 4 μC placed at the centre of the square.



1

OA = OB = OC = OD = r

Net force on charge $4\mu C$

$$\vec{F} = \vec{F}_{OA} + \vec{F}_{OB} + \vec{F}_{OC} + \vec{F}_{OD}$$ ½

$$\vec{F}_{OA} = -\vec{F}_{OC} \quad \Rightarrow \vec{F}_{OA} + \vec{F}_{OC} = 0$$

$$\vec{F}_{OB} = -\vec{F}_{OD} \quad \Rightarrow \vec{F}_{OB} + \vec{F}_{OD} = 0$$

$$\vec{F} = 0$$ ½

Alternatively

$$F_{OA} = F_{OC} = \frac{9\times10^{9}\times4\times10^{-6}\times1\times10^{-6}}{(15\sqrt{2}\times10^{-2})^{2}}$$

$$= 0.8\,N$$

½

$$F_{OB} = F_{OD} = 1.6\ N$$

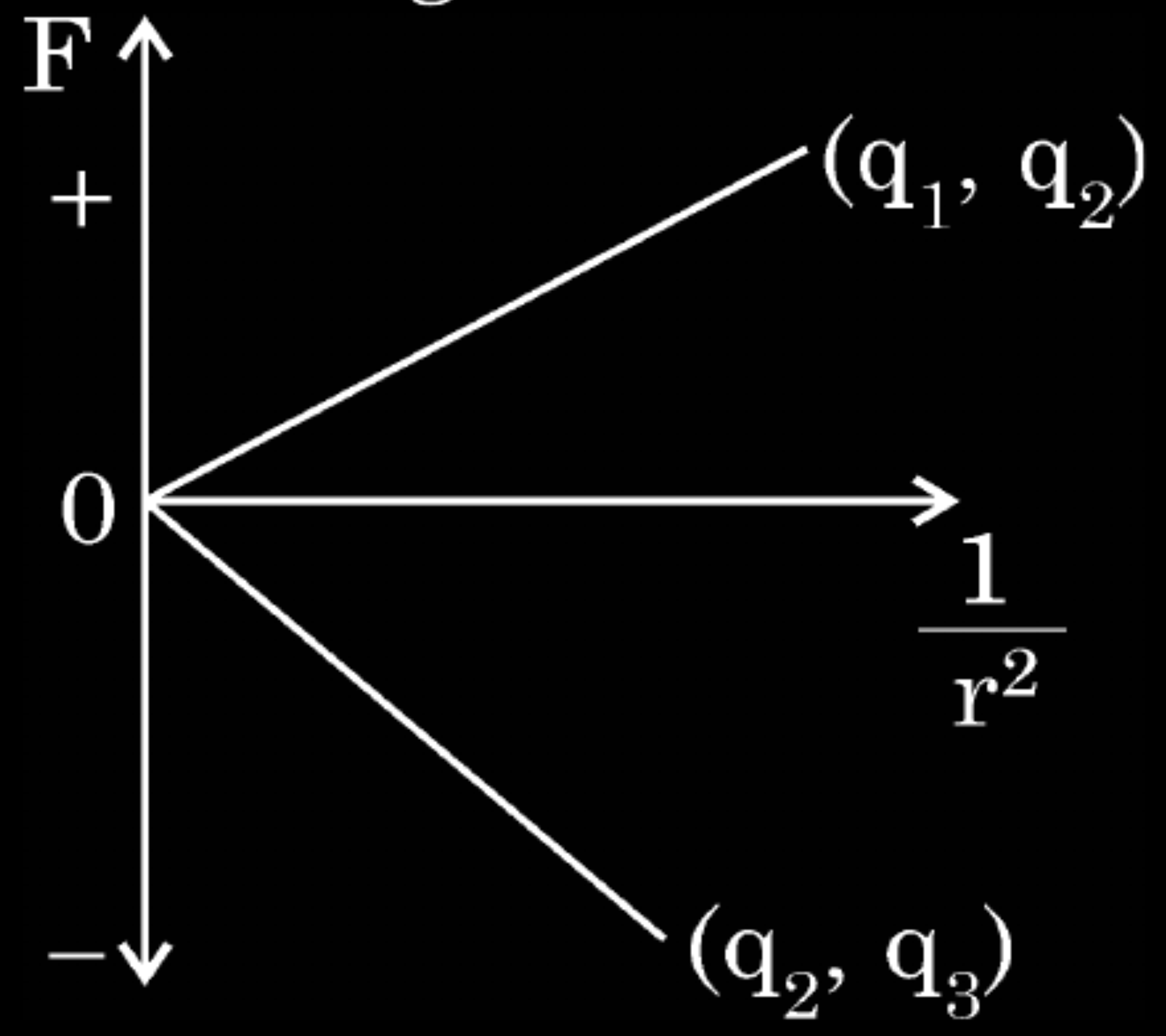
$$F_1 = F_{OA} - F_{OC} = 0$$

$$F_2 = F_{OB} - F_{OD} = 0$$

½

Net Force $F = 0$

2. The Coulomb force (F) versus ($1/r^2$) graphs for two pairs of point charges (q_1 and q_2) and (q_2 and q_3) are shown in figure. The charge q_2 is positive and has least magnitude. Then **1**



(A) $q_1 > q_2 > q_3$

(B) $q_1 > q_3 > q_2$

(C) $q_3 > q_2 > q_1$

(D) $q_3 > q_1 > q_2$

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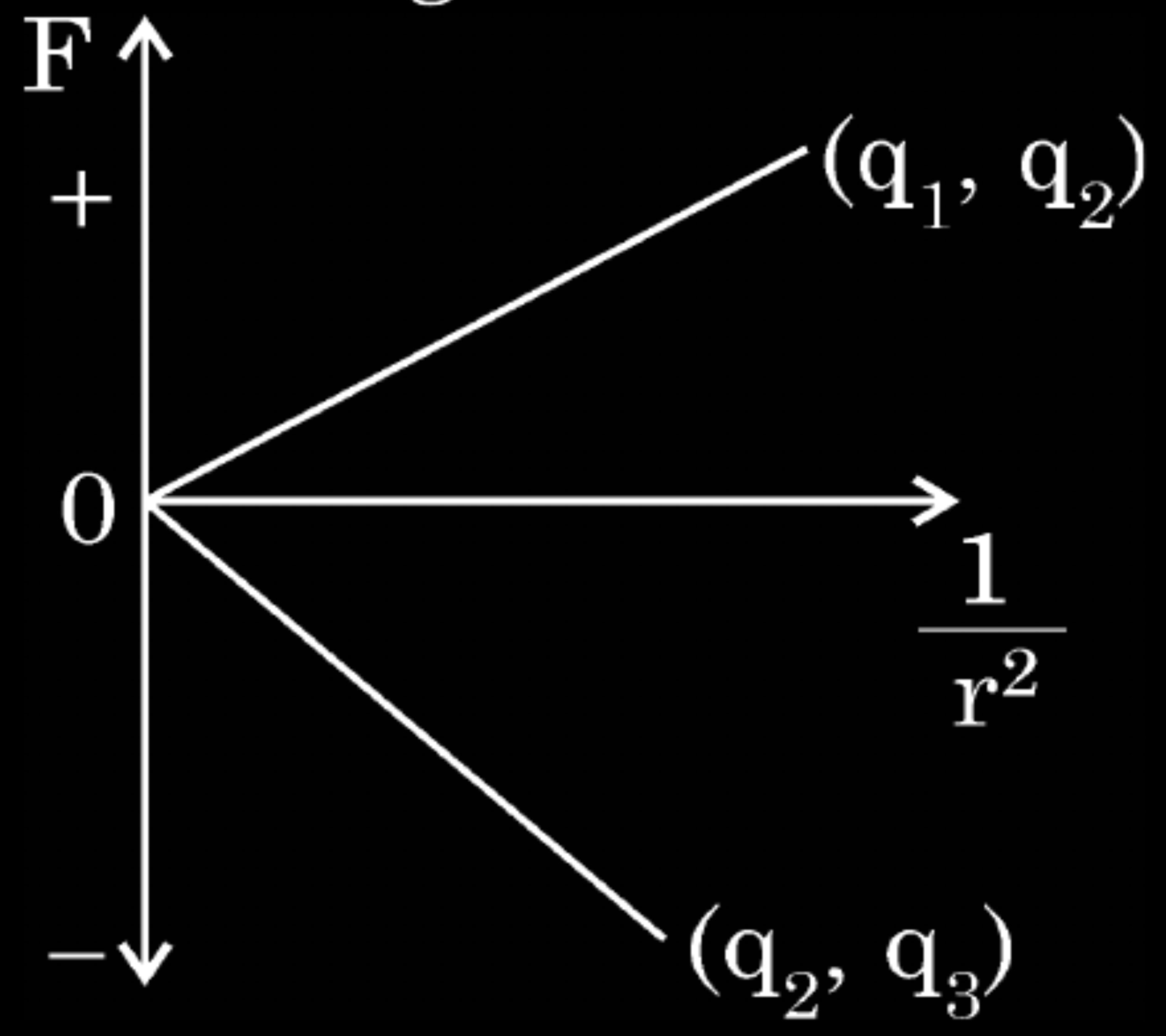


(A) $q_1 > q_2 > q_3$

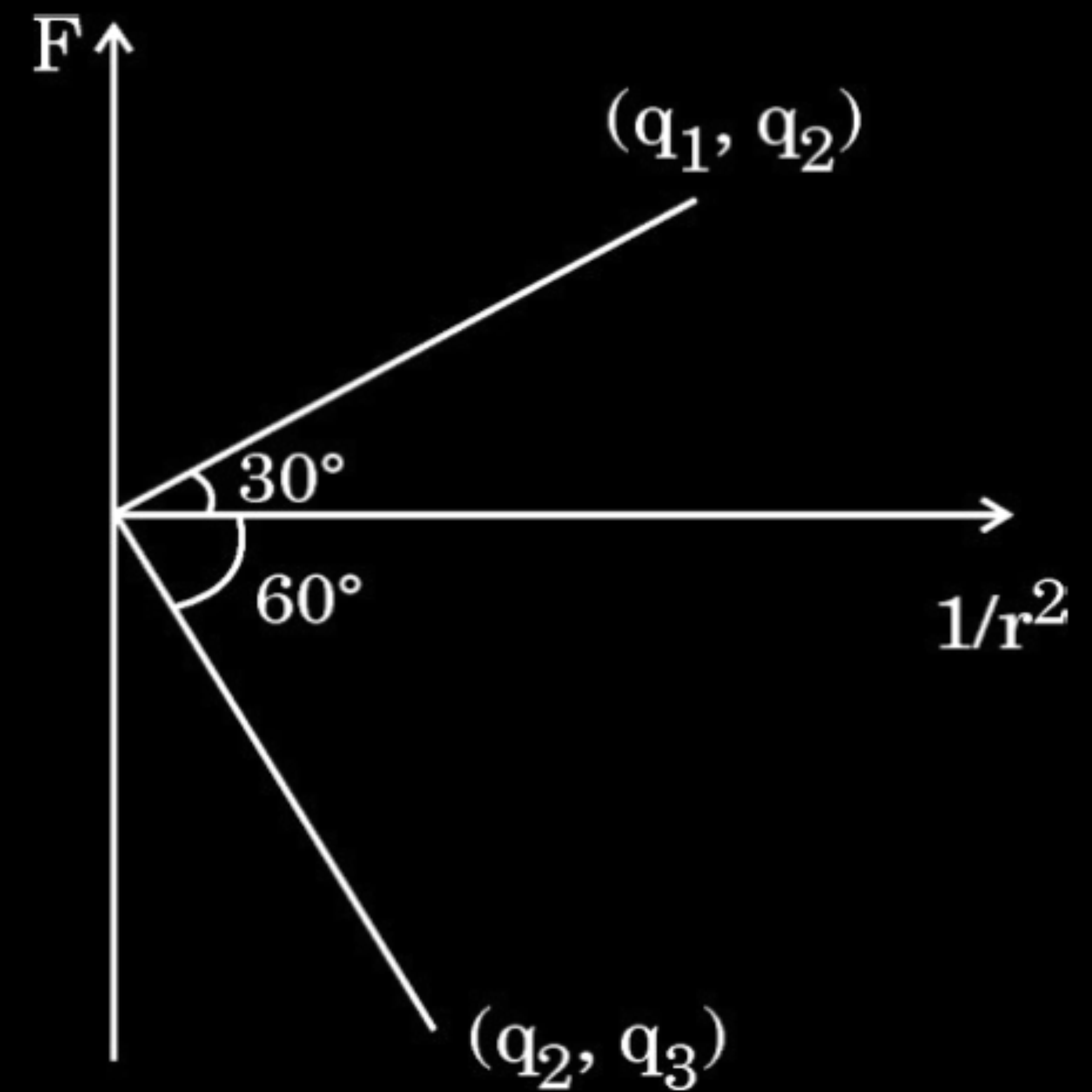
(B) $q_1 > q_3 > q_2$

(C) $q_3 > q_2 > q_1$

(D) $q_3 > q_1 > q_2$

(D) $q_3>q_1>q_2$

1. Coulomb force F versus $\left(\frac{1}{r^2}\right)$ graphs for two pairs of point charges (q_1 and q_2) and (q_2 and q_3) are shown in the figure. The ratio of charges $\left(\frac{q_1}{q_3}\right)$ is :



(A) $\sqrt{3}$

(B) $\frac{1}{\sqrt{3}}$

(C) 3

(D) $\frac{1}{3}$

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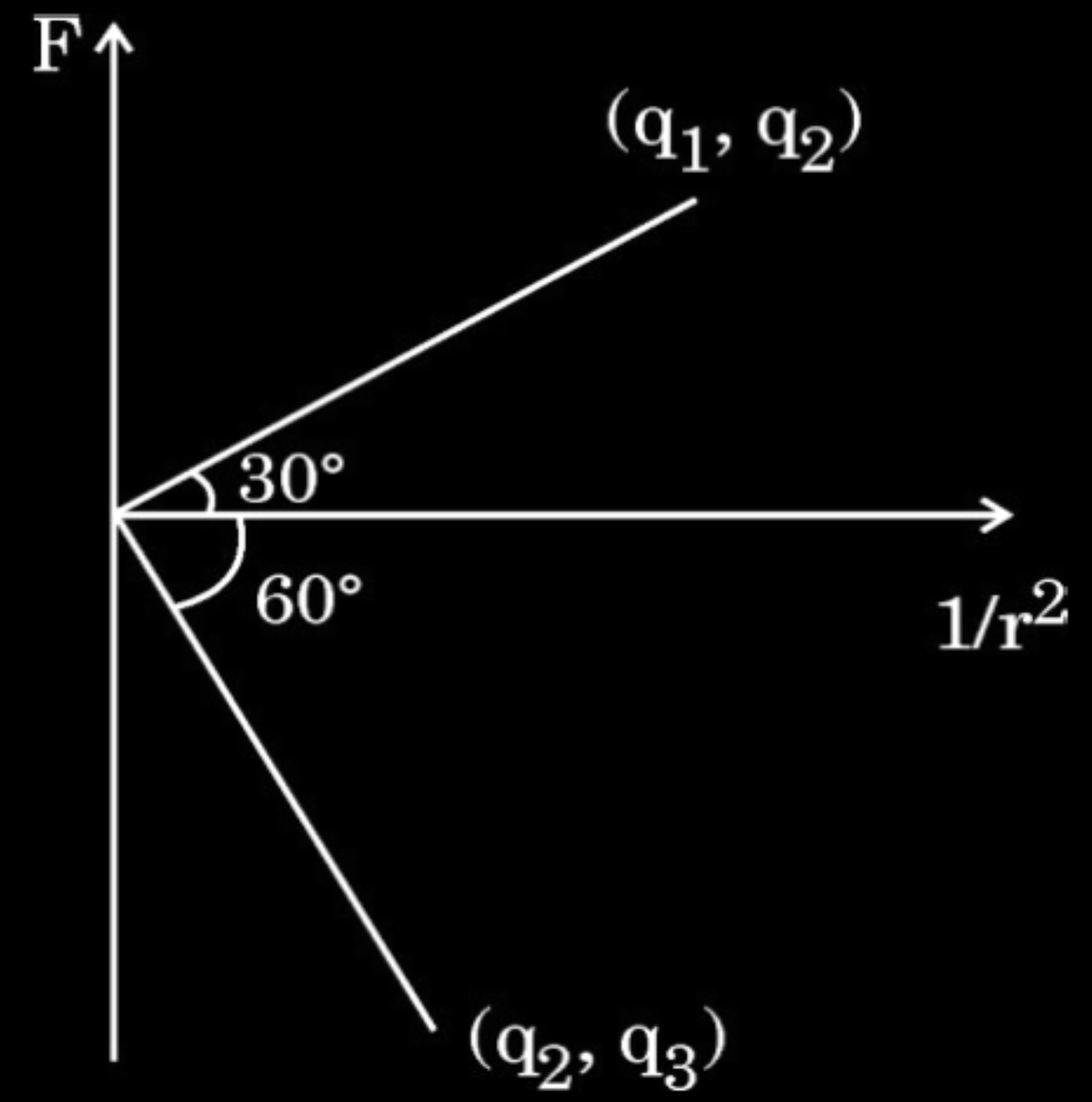

(A) $\sqrt{3}$

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(C) 3

(D) $\frac{1}{3}$

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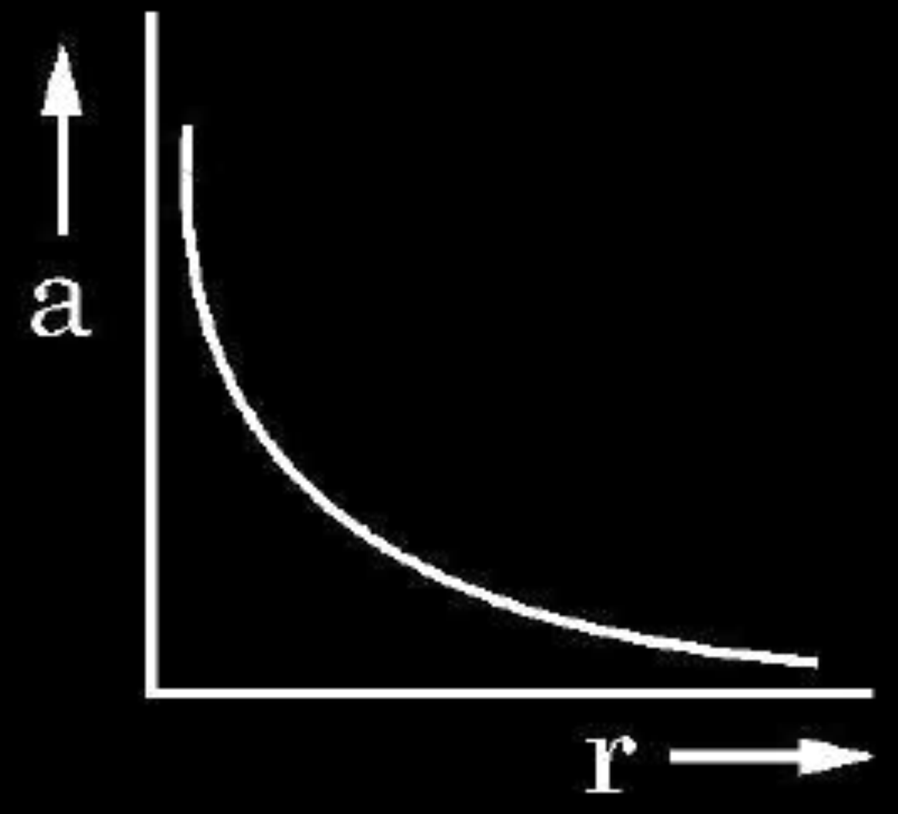
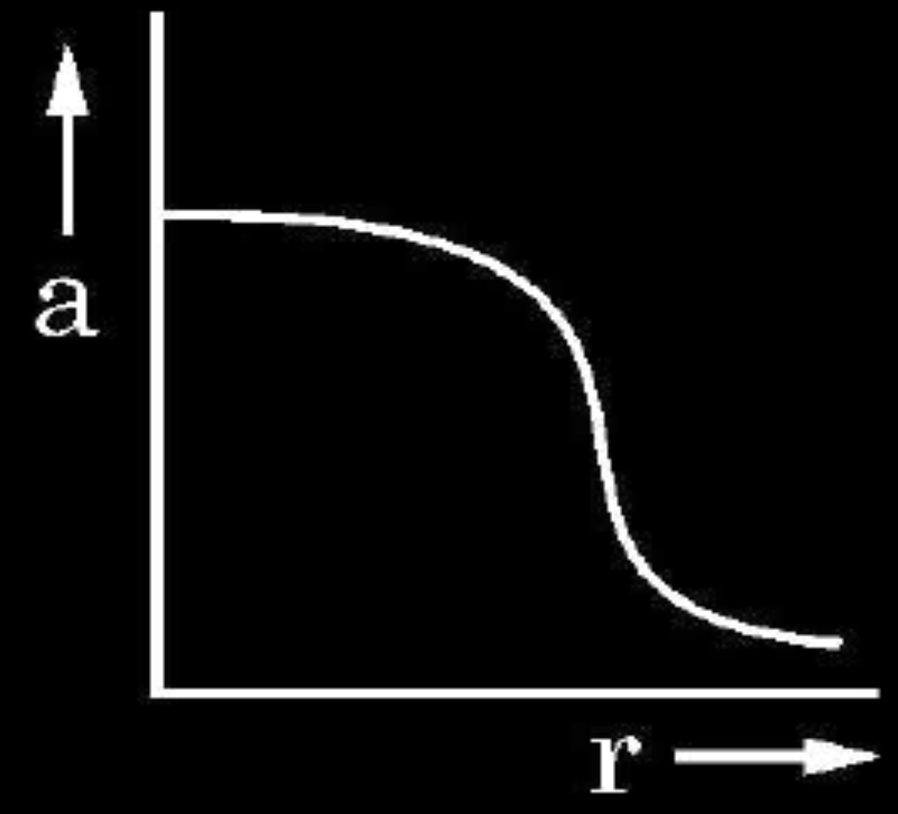
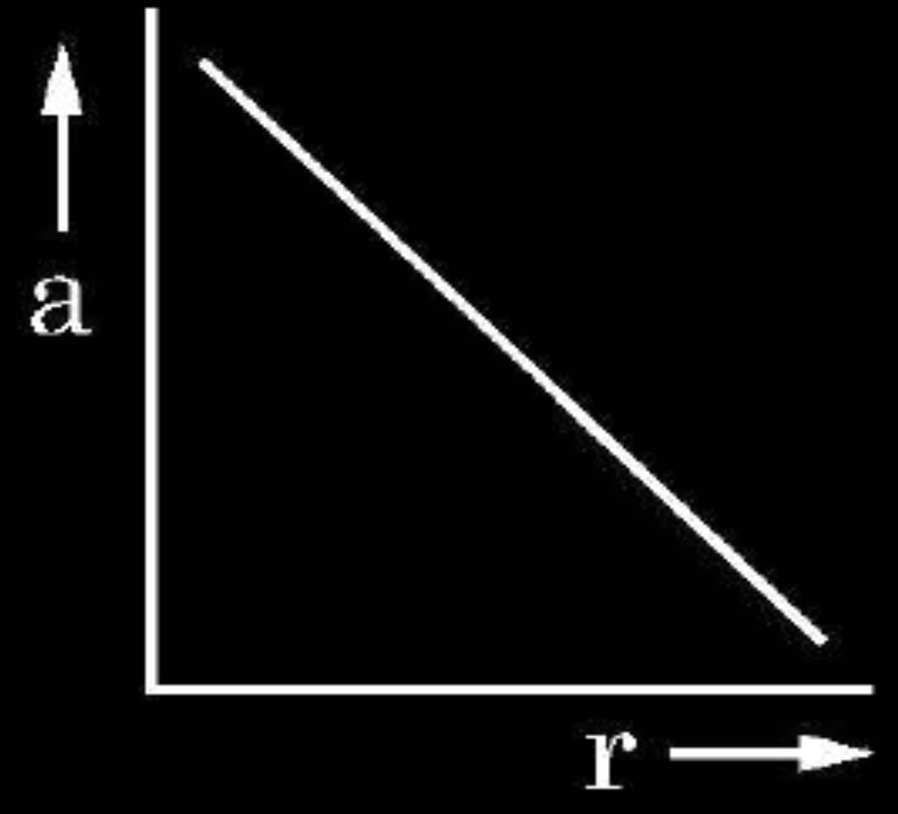
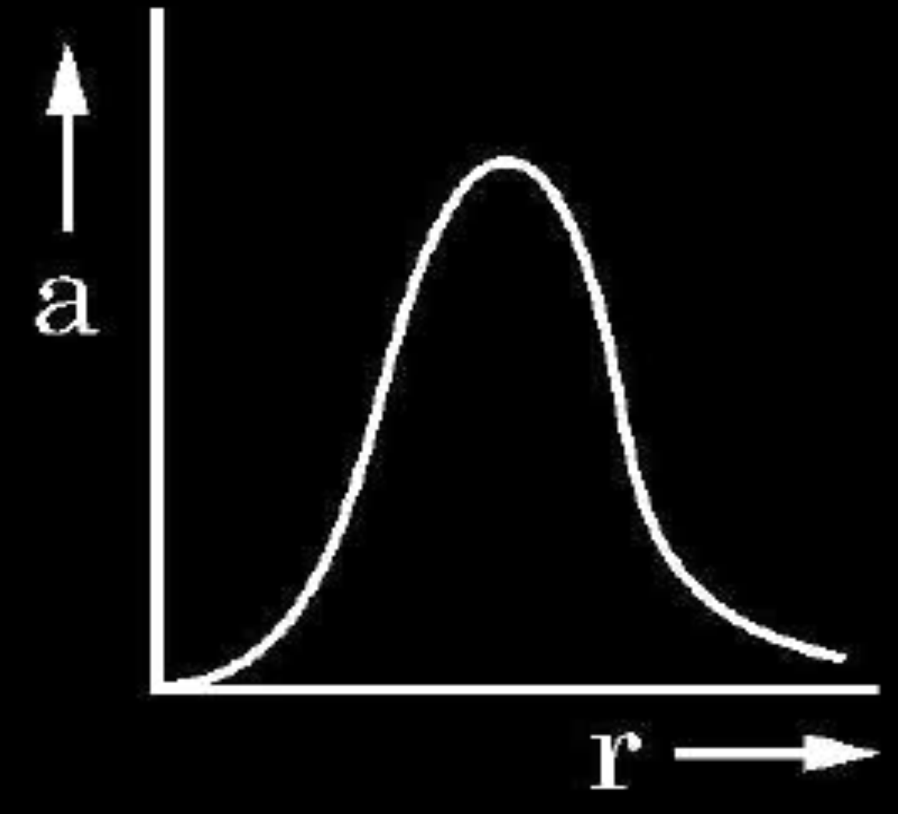
1. A charge Q is fixed in position. Another charge q is brought near charge Q and released from rest. Which of the following graphs is the correct representation of the acceleration of the charge q as a function of its distance r from charge Q ? **1**

(A) 

(B) 

(C) 

(D) 

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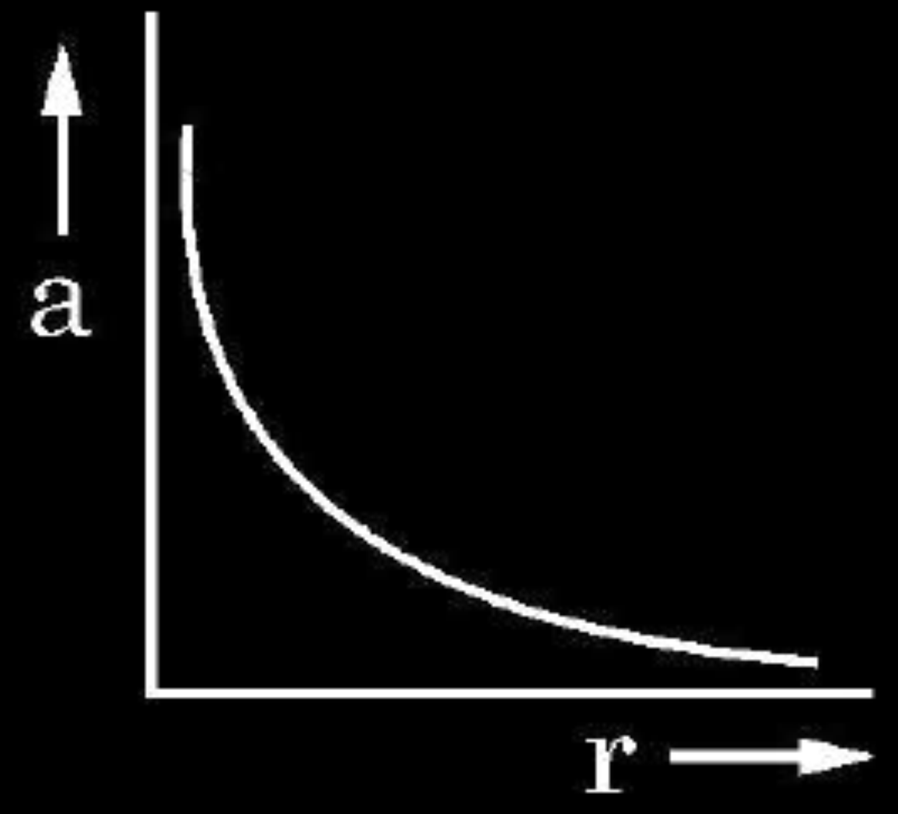
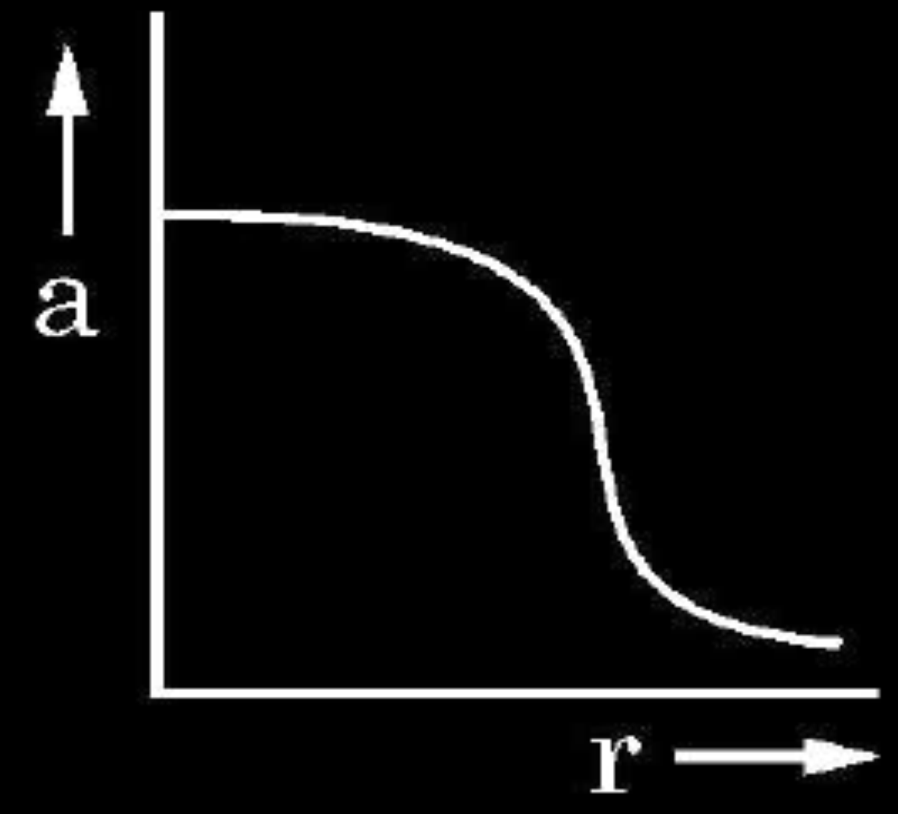
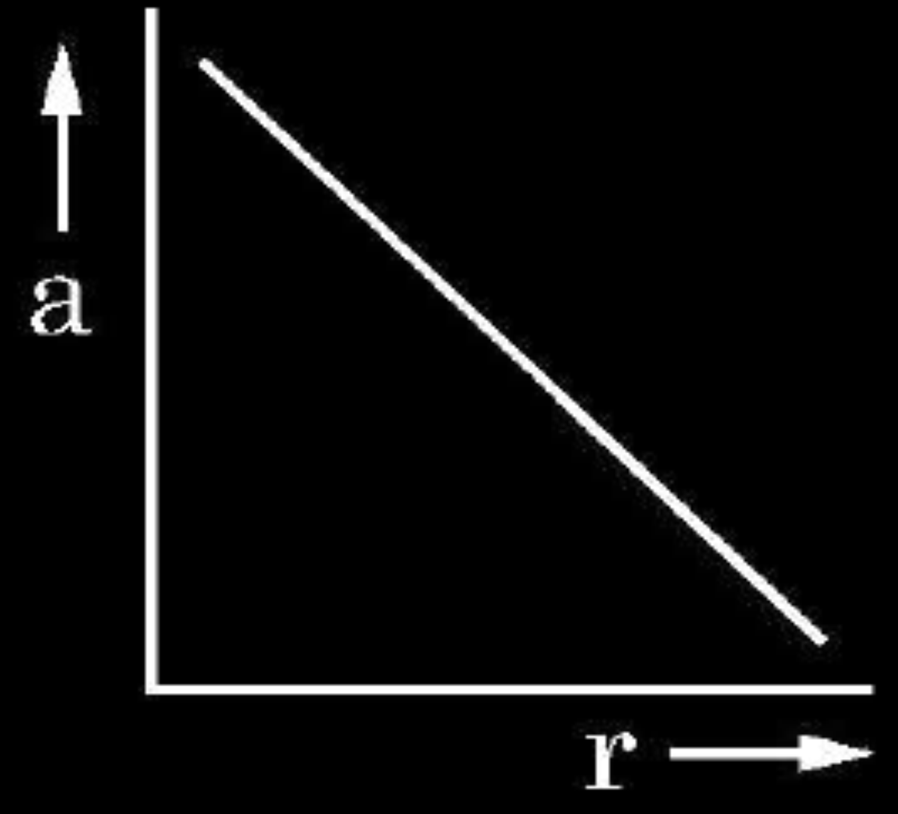
(A)



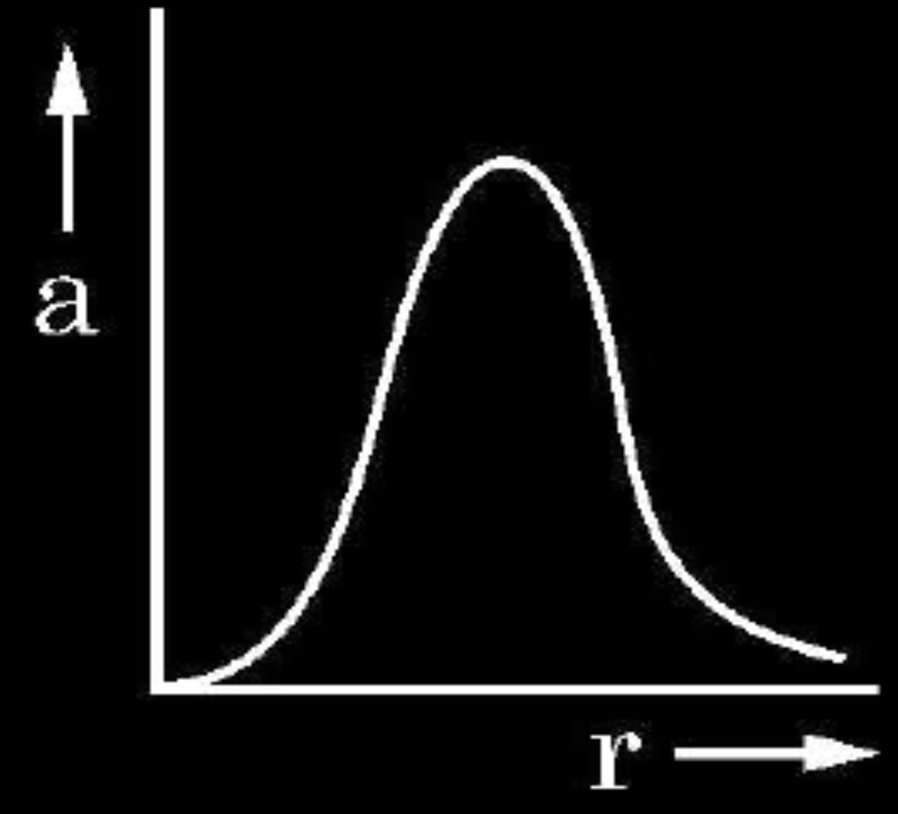
(B)



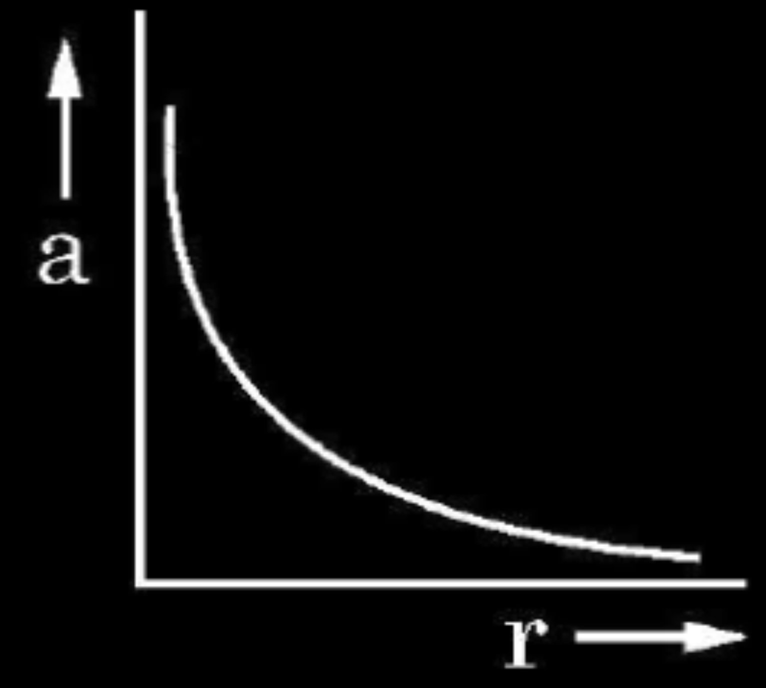
(C)



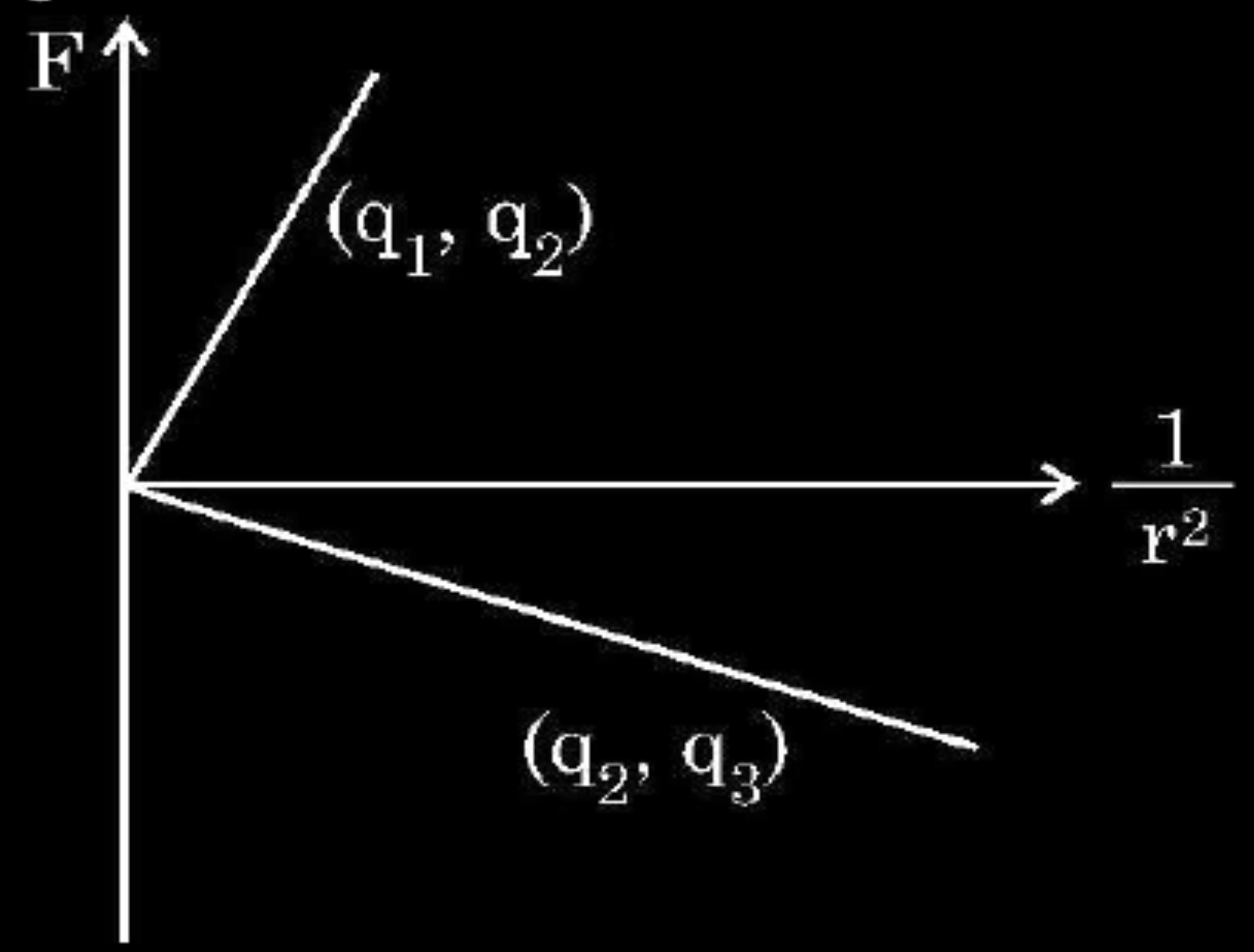
(D)



(A)



1. Figure shows variation of Coulomb force (F) acting between two point charges with $\frac{1}{r^2}$, r being the separation between the two charges (q_1, q_2) and (q_2, q_3). If q_2 is positive and least in magnitude, then the magnitudes of q_1, q_2 and q_3 are such that **1**



(A) $q_2 < q_3 < q_1$

(B) $q_3 < q_1 < q_2$

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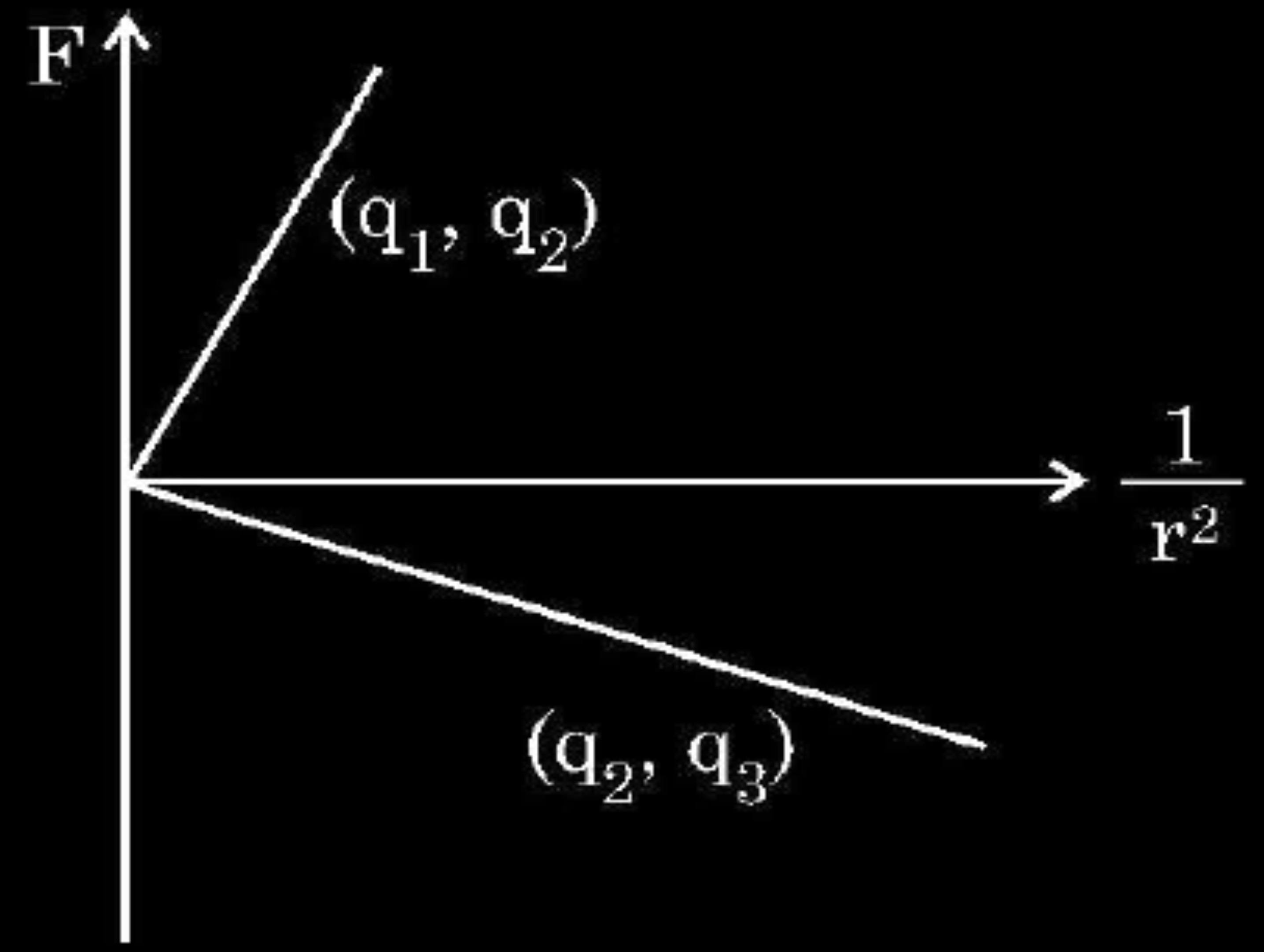
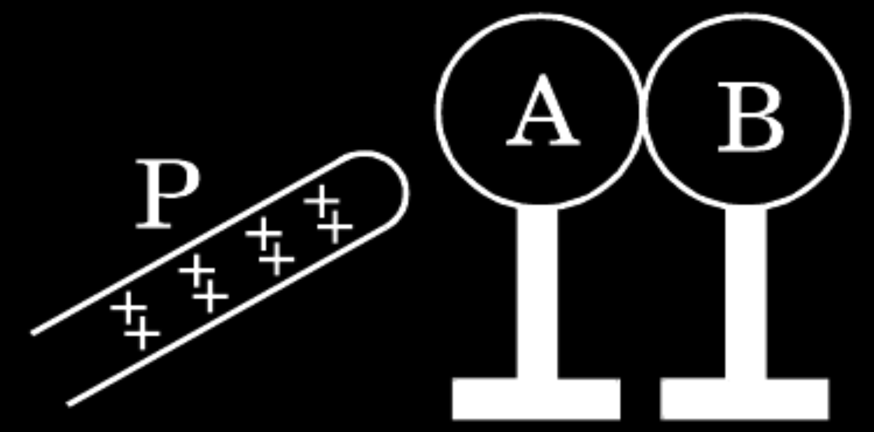


(A) $q_2 < q_3 < q_1$ (B) $q_3 < q_1 < q_2$

(C) $q_1 < q_2 < q_3$ (D) $q_2 < q_1 < q_3$

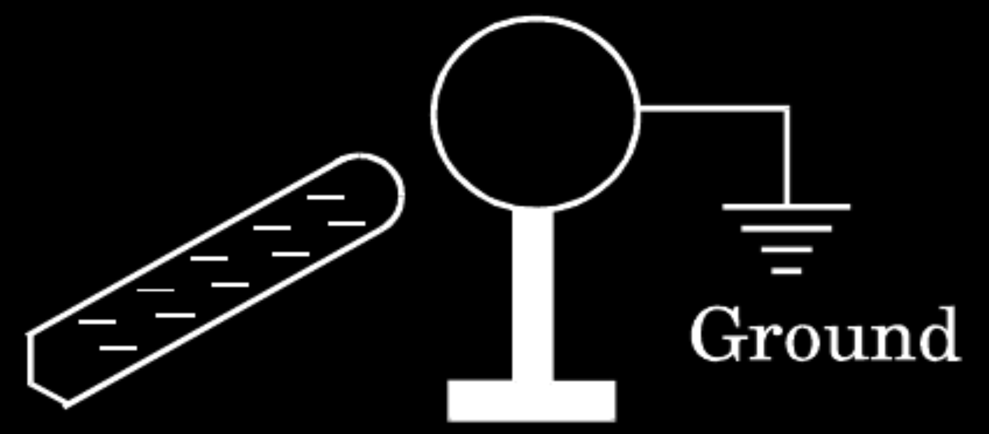
(A) $q_2 < q_3 < q_1$

Two metallic spheres A and B kept on insulating stands are in contact with each other. A positively charged rod P is brought near the sphere A as shown in the figure. The two spheres are separated from each other, and the rod P is removed. What will be the nature of charges on spheres A and B ?



OR

A metal sphere is kept on an insulating stand. A negatively charged rod is brought near it, then the sphere is earthed as shown. On removing the earthing, and taking the negatively charged rod away, what will be the nature of charge on the sphere ? Give reason for your answer.



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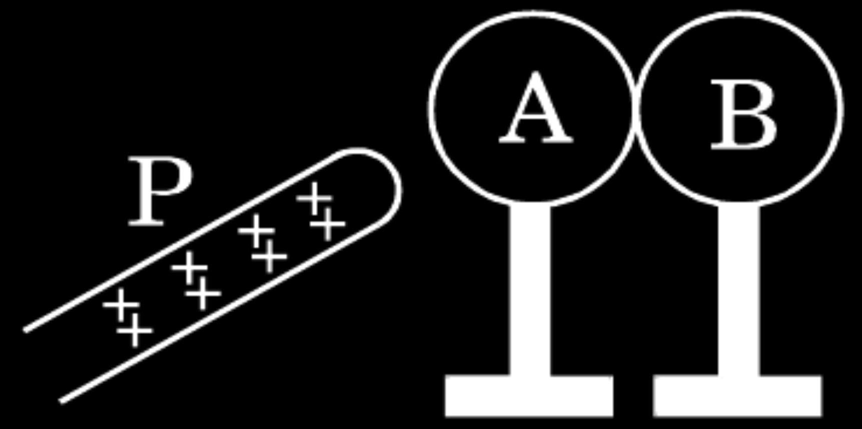

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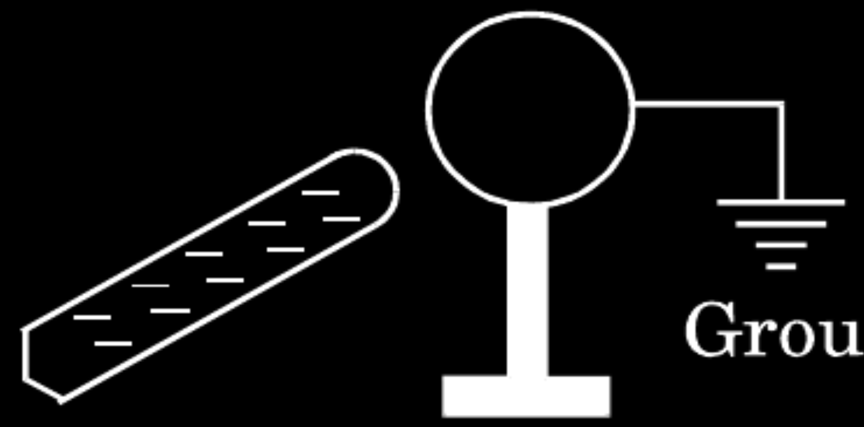
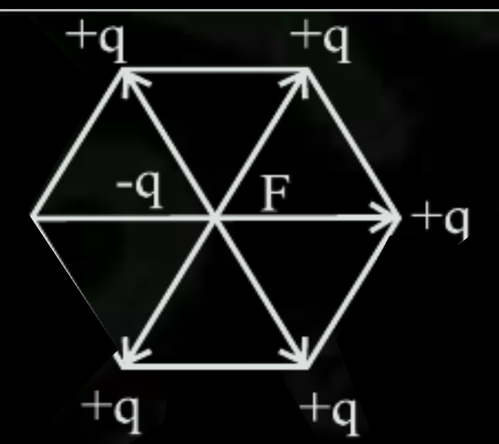


Sphere A will be negatively charged	½
Sphere B will be positively charged	½
Alternatively- B will be similarly charged to the rod and A will be oppositely charged.	½+½
OR	
Sphere will be positively charged.	½
Reason - Electrostatic Induction	½

Five point charges, each of charge $+q$ are placed on five vertices of a regular hexagon of side 'l'. Find the magnitude of the resultant force on a charge $-q$ placed at the centre of the hexagon.



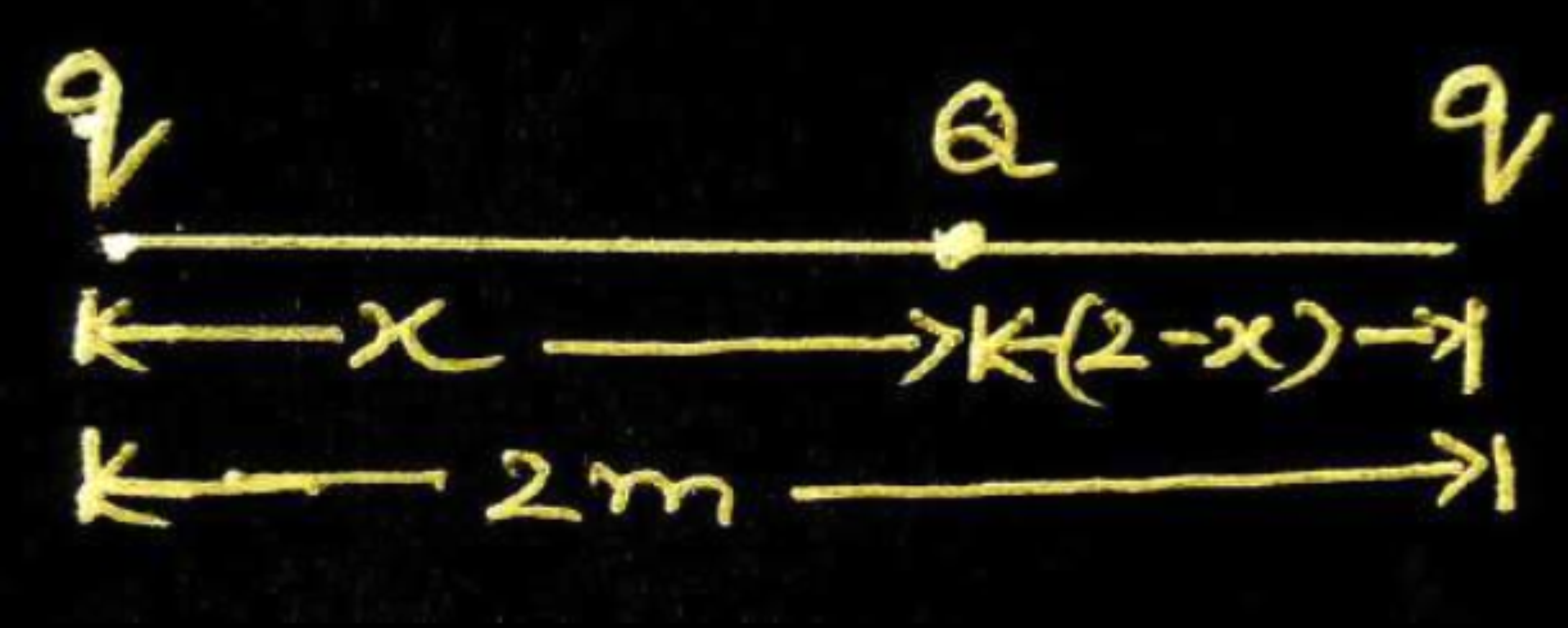
Alternatively

The forces due to the charges placed diagonally opposite at the vertices of hexagon, on the charge -q cancel in pairs. Hence net force is due to one charge only. 1

Net Force $|\overline{F}| = \frac{1}{4\pi\varepsilon_0}\ \frac{q^2}{l^2}$ 1

(b) Two identical point charges, q each, are kept 2m apart in air. A third point charge Q of unknown magnitude and sign is placed on the line joining the charges such that the system remains in equilibrium. Find the position and nature of Q.

(b)



½

System is in equilibrium therefore net force on each charge of system will be zero.

For the total force on 'Q' to be zero

$$\frac{1}{4\pi \in_0}\frac{qQ}{x^2} = \frac{1}{4\pi \in_0}\frac{qQ}{(2-x)^2}$$ ½

x = 2 – x ½

2x = 2
x = 1 m

(Give full credit of this part, if a students writes directly 1m by observing the given condition)

For the equilibrium of charge "q" the nature of charge Q must be opposite to the nature of charge q. ½